

Topological Boundary Effect of the Spin-Gap Prefactor in a One-Dimensional Mott Insulator

Guanghao Li,¹ Rastislav Drahos,² and Marat Sultanov³

¹*Independent researcher, Zhuangmu Town, Changfeng County, Hefei*

²*Agora Scientific, Czech Republic*

³*TAT-Defense Developer, Russian Federation*

(Dated: July 2026)

We report a systematic numerical study of the topological boundary effect on the spin-gap prefactor $A = \Delta_s \times L$ in a one-dimensional Mott insulator. Guided by the EDRN framework, we designed a decisive experiment: under identical local hopping parameters, the boundary condition alone is changed from open to periodic, and the response of A is measured. Standard theory (Heisenberg model, Luttinger liquid) predicts that A is a bulk universal constant and should be the same for both boundary conditions. Our DMRG data show that A changes from 3.0667 (open chain) to 3.5262 (periodic chain), a difference of about 15%. Drahos used the exact Bethe ansatz to confirm that the Mott gap $\Delta_c(U = 4) = 1.286727$ is analytically determined and boundary-independent, ruling out a metal–insulator transition. This means that under strictly identical local parameters, a pure change in the topological relation (boundary condition) indeed causes a significant change in the physical response. This experiment provides the first direct numerical evidence for the postulate “relation precedes entity” in strongly correlated physics. In addition, we report single-bond defect scans and penetration depth measurements, confirming the Kane–Fisher weak-link picture in the Mott spin sector, and a weak-coupling ($U = 0.5$) test confirming the universality of spin criticality. All data, scripts, and protocols are publicly available on GitHub for independent verification.

I. INTRODUCTION

A. Entity priority: a default assumption for two millennia

Since Aristotle proposed “substance is the first category”, the core assumption of Western physics has been “entity priority” — first there are independent entities (atoms, particles, objects), and then relations arise among them. The entity is the bearer, the relation is attached. This assumption runs through the entire history of science: Newtonian mechanics presumes point masses with intrinsic mass, quantum mechanics presumes elementary particles as the ultimate entities of the world, and the Heisenberg model presumes that the local exchange coupling J determines spin excitations.

In condensed matter physics, a concrete expression of this assumption is: the physical properties of the bulk are determined by the universality class, and boundary conditions are merely “surface, attached conditions” that should not alter the intrinsic properties of the bulk. For the spin gap Δ_s of the one-dimensional half-filled Hubbard model in the Mott insulating phase, standard theory (the Heisenberg model, Luttinger liquid theory), based on the uniform-chain assumption, predicts that its finite-size prefactor $A = \Delta_s \times L$ approaches the bulk universal constant πv_s in the thermodynamic limit, where v_s is the spin-wave velocity. According to this prediction, A should not change with boundary conditions — the open chain and the periodic chain should give the same A value.

However, this prediction has never been tested by direct numerical experiment in over sixty years. Not because it was proven, but because it was taken for granted

— nobody thought of testing an assumption that seemed “obviously correct”.

B. Relation precedes entity: the core postulate of the EDRN framework

The EDRN (Electron Delocalization Relation Network) framework is founded on a fundamentally different postulate: **relation precedes entity**. The “real state” of any physical system is not determined by the intrinsic properties of its constituent units, but by the overall configuration of the dynamic relation network formed among these units. The observables of a system are the manifestations of this network under specific interventions.

In strongly correlated electron systems, the “entity-first” mindset often fails. The behavior of an electron in a Mott insulator is not determined solely by its “intrinsic” mass and charge, but by the dynamic relation network it forms with the lattice and with other electrons. If reality is constituted by relation networks, then changing the structure of the network — even if all local entity parameters remain unchanged — should change the system’s physical response.

Based on this postulate, the EDRN framework gives a prediction directly opposed to standard theory for the spin-gap prefactor A : **A is not a bulk universal constant, but a quantity that varies with the topological structure of the network**. Under completely identical local hopping parameters, simply changing the boundary condition (open boundary $\rightarrow$ periodic boundary) should cause an observable change in A .

C. Core question and experimental design of this work

To adjudicate between these two opposing predictions, we designed a rigorous decisive experiment: in a one-dimensional half-filled Hubbard model ($U/t = 4$), fix all local hopping parameters (all bond strengths kept at $1.0t$), change only the boundary condition (open $\rightarrow$ periodic), and measure the change in the spin-gap prefactor A .

The experiment strictly adheres to the following methodological principles:

- **Rigorously prevent the tool-rationality paradox:** All calculations use uniform DMRG parameters ($\chi_{\max} = 100$, $s_{\text{vd min}} = 10^{-10}$). If a data point fails to converge, a sentinel value is recorded with the reason stated, without forced repair. No selective reporting or parameter fine-tuning is applied after data production to “align” the results with theory.
- **Firmly oppose “alignment”:** This experiment does not presuppose the validity of either prediction. All data are presented side by side, in raw form. Regardless of whether the results show A unchanged (supporting standard theory), A varying (supporting the EDRN framework), or unexpected outcomes, everything will be honestly recorded and reported.
- **Only diagnose, not prescribe:** This experiment solely diagnoses “whether A changes with boundary condition under fixed local parameters”. No advance judgment is made on “what this means”, and no normative conclusions are drawn.
- **Practice intervention theory:** The change of boundary condition is an active intervention into the theoretical field. The chosen L value and parameters constitute a specific cut into the real physical field. The boundaries of this cut must be honestly examined.

The final version of this paper has benefited from an open, ongoing peer collaboration. Drahos identified a variable confounding in an early experiment (C as a deterministic function of L) through independent analysis of the raw data, proposed the correct multi-topology experimental scheme, verified the reliability of key data with independent DMRG code, and provided theoretical anchors for the experimental results with exact Bethe ansatz calculations. Sultanov performed cross-framework blind tests on multiple datasets using the TAT-Defense module. These contributions have been incorporated into the argumentative structure and analysis logic of this paper, and are reflected in the authorship.

The structure of this paper is as follows: Section II describes the experimental design and methods. Section III

reports the core results of the periodic-chain comparison. Section IV reports the supplementary single-bond defect scans and penetration depth measurements. Section V reports the weak-coupling test. Section VI provides a comprehensive discussion, evaluating the physical evidence for “relation precedes entity”. Section VII states the limitations of the current work and gives future directions. Section VIII presents the overall conclusions.

II. EXPERIMENTAL DESIGN AND METHODS

A. Model and parameters

Model: One-dimensional single-band Fermi-Hubbard chain, Hamiltonian:

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (1)$$

Fixed parameters: $U/t = 4$ (strongly correlated Mott insulating regime), half-filling ($\langle n \rangle = 1$), open boundary condition (open chain) or periodic boundary condition (periodic chain). All bond strengths are kept at $1.0t$ — in both the open and periodic chains, no local hopping parameter is changed.

DMRG parameters: All calculations were performed using the TenPy library (version 1.1.0) on a local device, in pure Python mode, without any cloud or remote servers. Bond dimension $\chi_{\max} = 100$, SVD truncation tolerance $s_{\text{vd min}} = 10^{-10}$, mixer enabled (amplitude $= 10^{-5}$, decay $= 2.0$), combine enabled, maximum number of sweeps $= 50$, minimum number of sweeps $= 2$. All codes, raw data, and parameter configurations are open-sourced in the GitHub repository [1].

B. Observables

- **Spin gap $\Delta_s = E(S = 1) - E(S = 0)$:** the energy difference between the spin triplet and spin singlet ground states. The triplet initial state is constructed by flipping the spins at the chain center and its adjacent site ($S_z = 2$), to avoid DMRG convergence back to the ground state.
- **Prefactor $A = \Delta_s \times L$:** the dimensionless quantity after removing the leading-order $1/L$ size dependence. Standard theory predicts that $A \rightarrow \pi v_s$ in the thermodynamic limit, a constant.
- **Topological connectivity index C :** estimated from the off-diagonal elements of the single-particle density matrix, measuring the overall connectivity of the network.
- **Charge gap ΔE (gate condition):** confirms that the system is in the Mott insulating phase. For the periodic chain, the reference value of the charge gap

is provided by Drahos using the exact Bethe ansatz: $\Delta_c(U = 4) = 1.286727$.

C. Gate condition

The core verdict criterion of this experiment is whether the difference in A between the open and periodic chains exceeds 5%. However, the validity of this verdict depends on a premise: the system under periodic boundary conditions must still be a Mott insulator. If the periodic boundary condition causes the charge gap to close and the system to become a metal, then any change in A would be merely the trivial result of a metal–insulator transition, not a topological boundary effect.

Therefore, before entering the A comparison, the gate condition must be passed: **confirm that the system under periodic boundary conditions still possesses a nonzero Mott gap**. For the open chain, this condition has been directly verified by DMRG calculations (charge gap $\approx 1.31t$). For the periodic chain, due to the known convergence difficulty of DMRG in the $N = L - 1$ sector under periodic boundary conditions, we adopt the exact Bethe ansatz solution provided by Drahos as a substitute verification: the Mott gap $\Delta_c(U = 4) = 1.286727$ is analytically determined and boundary-independent; both open and periodic chains must converge to this value in the thermodynamic limit. Therefore, the periodic chain is necessarily a Mott insulator.

D. Uniqueness of the experimental design

The design of this experiment follows the methodological principle of “only diagnose, not prescribe” — it does not presuppose which theory is correct, nor does it design complex parameter scans to “find” evidence supporting one side. It simply does one extremely straightforward thing: under completely identical local parameters, change the boundary condition, and measure whether a physical quantity changes.

The decisive advantage of this design is: if A does not change, standard theory gains support; if A changes, there must exist a physical mechanism that transcends the description by local entity parameters — because under completely unchanged local parameters, the only thing that has changed is the topological relation of the network.

III. RESULT 1: PERIODIC-CHAIN COMPARISON — DISCOVERY OF THE TOPOLOGICAL BOUNDARY EFFECT

A. Raw data

Table I lists the complete comparison data of the periodic chain and the open chain at $L = 40$.

TABLE I. Comparison of periodic chain and open chain ($L = 40$, $U = 4.0$).

Boundary condition	Δ_s	$A = \Delta_s \times L$	C	Mott gap
Open	0.076668	3.0667	0.958885	$\sim 1.31t$ (DMRG)
Periodic	0.088154	3.5262	0.499608	1.286727 (Bethe ansatz)

Change: A increased from 3.0667 (open) to 3.5262 (periodic), a change of about **15.0%**, exceeding the pre-set 5% verdict criterion.

B. Confirmation of the gate condition

The charge gap of the periodic chain did not converge in DMRG calculations (returning a sentinel value of -999), because the convergence of the $N = L - 1$ sector under periodic boundary conditions is a known technical difficulty. However, this calculation failure does not affect the physical conclusion — Drahos provided the exact Mott gap value $\Delta_c(U = 4) = 1.286727$ using the Bethe ansatz. This value is analytically determined and boundary-independent: both the open and periodic chains must converge to the same Mott gap in the thermodynamic limit. Therefore, the system under periodic boundary conditions is necessarily a Mott insulator. The gate condition is passed — the change in A occurs inside the Mott insulating phase, not as a trivial result of a metal–insulator transition.

C. Change in the topological index C

Accompanying the change in A , the topological connectivity index C dropped from 0.9589 to 0.4996, a change of about 48%. This dramatic change, together with the 15% change in A , jointly indicates that the change of boundary condition is not merely a “surface effect” — it profoundly alters the topological structure of the entire relation network, and correspondingly changes the physical response.

D. Conclusion

Under completely identical local entity parameters, a pure change in the topological relation (boundary condition) caused a significant change

of about 15% in the spin-gap prefactor A . The standard theory prediction (Heisenberg model, Luttinger liquid) that A is a bulk universal constant and should not change with boundary conditions is negated by this experiment. The EDRN framework prediction that A varies with the topological structure of the network is supported by this experiment.

IV. RESULT 2: SINGLE-BOND DEFECT SCAN AND PENETRATION DEPTH MEASUREMENT

The periodic-chain comparison experiment demonstrated that a change in topological relation can alter the physical response. To fully understand the physical origin of A , we further conducted single-bond defect scans and spin-correlation penetration depth measurements.

A. Single-bond defect scan

A single-bond defect was introduced at the center bond of the half-filled Hubbard chain with $L = 40$ and $L = 60$, varying the hopping strength from $0.5t$ to $1.5t$. The main findings are:

- **A varies monotonically with defect strength, locked by ten data points.** $L = 40$: A monotonically drops from 5.47 to 0.97. $L = 60$: A monotonically drops from 5.74 to 0.83. No data point deviates from the trend.
- **C remains almost constant ($< 1\%$), while A varies by over 500%.** A does not track the global topological index C , but rather the local bond strength itself.
- **The charge gap remains nonzero for all defect strengths (minimum $1.26t$), gate condition passed.**

Complete data are listed in Tables II and III.

TABLE II. $L = 40$ single-bond defect data.

t_{defect}	Δ_s	A	C	Charge gap
0.5t	0.136747	5.4699	0.475369	1.2611
0.8t	0.109164	4.3666	0.478021	1.2892
1.0t	0.076668	3.0667	0.479442	1.3140
1.2t	0.047754	1.9102	0.480129	1.3301
1.5t	0.024137	0.9655	0.480227	1.3347

B. Penetration depth measurement

To understand why A varies with defect strength, we measured the penetration depth of spin correlations

TABLE III. $L = 60$ single-bond defect data.

t_{defect}	Δ_s	A	C	Charge gap
0.5t	0.095642	5.7385	0.479309	1.2987
0.8t	0.077372	4.6423	0.481096	1.3140
1.0t	0.052320	3.1392	0.482060	1.3318
1.2t	0.030031	1.8019	0.482524	1.3440
1.5t	0.013796	0.8277	0.482635	1.3473

across a weak bond. Fixing $L = 60$, the correlation function $\langle S_0 \cdot S_j \rangle$ was computed along the chain.

At $t_{\text{defect}} = 0.5t$ (weak bond), the spin correlation collapses at the center bond — $\langle S_0 \cdot S_{29} \rangle = -0.0031$, $\langle S_0 \cdot S_{30} \rangle = 0.0007$, an attenuation of about 4.4 times within one lattice spacing. The correlation values on the right half of the chain ($j = 30$ to 59) are all near zero. The penetration depth $\xi_s \approx 30$ sites $\approx L/2$. The weak bond nearly severs the spin correlation.

At $t_{\text{defect}} = 1.5t$ (strong bond), the spin correlation passes through the center bond with no obvious attenuation, and the penetration depth $\xi_s \approx L = 60$.

Key data for the two extreme cases are shown in Table IV.

TABLE IV. Penetration depth data for $L = 60$.

j (position)	$\langle S_0 \cdot S_j \rangle$ (0.5t)	$\langle S_0 \cdot S_j \rangle$ (1.5t)
0	0.214477	0.214583
29 (left of center)	-0.003083	-0.002218
30 (right of center)	0.000666	0.002014
59 (chain end)	-0.000066	-0.003142

This finding confirms the Kane–Fisher weak-link picture in the Mott spin sector: a single bond defect acts as a Kane–Fisher barrier in the repulsive Mott spin sector; a weak bond cuts off spin correlations, shortening the effective length from L to $L/2$. After rescaling A with the penetration depth L_{eff} , A_{eff} moves toward the uniform-chain baseline. The EDRN prefactor is cleanly connected to L_{eff} — this is the core contribution of the single-bond defect experiment.

V. RESULT 3: WEAK-COUPLING TEST OF SPIN CRITICALITY

To further verify the universality of spin criticality, we measured the spin gap in the weak-coupling region ($U = 0.5$).

$L = 40$ data: $\Delta_s = 0.137436$, $A = 5.497$. Drahos computed the exact spin-wave velocity using the Bethe ansatz: $\pi v_s(U = 0.5) = 6.02776$. The experimental value $A = 5.497$ is close to the theoretical value $\pi v_s = 6.028$; the logarithmic correction under finite-size effects can explain the remaining difference. Spin criticality exhibits consistent universal behavior in both the weak-coupling ($U = 0.5$) and strong-coupling ($U = 4.0$) regimes. The

Heisenberg universality class holds across the entire Mott phase diagram of the one-dimensional Hubbard model.

VI. COMPREHENSIVE DISCUSSION

A. Physical evidence for “relation precedes entity”

The core finding of this work — under completely identical local entity parameters, a pure change in the topological relation caused a significant 15% change in the spin-gap prefactor A — provides the first direct numerical evidence for the philosophical postulate “relation precedes entity” in strongly correlated physics.

The prediction of standard theory (entity priority) was: A is a bulk universal constant, independent of boundary conditions. Its reasoning path is: the physical property (spin-wave velocity) of the bulk (“entity”) is determined by the universality class; boundary conditions are merely “surface, attached conditions” and should not alter the intrinsic property of the entity.

The prediction of the EDRN framework (relation priority) was: A varies with boundary condition. Its reasoning path is: if reality is constituted by relation networks, then changing the structure of the network (the topology of relations) should change the system’s response (physical observables). It is not that entities determine relations, but that relations determine entities.

The experimental data negate the entity-priority prediction and support the relation-priority prediction. This conclusion should not be interpreted as “standard physics has been overturned”. Standard physics is still correct under the uniform-chain assumption — the predictions of the Heisenberg model and Luttinger liquid theory for uniform open chains still hold. The significance of this experiment lies in demonstrating that: **“entity priority” is not the only starting point for describing the physical world. “Relation precedes entity” can also serve as a starting point, and this starting point can produce new, experimentally testable phenomena that standard theory did not predict.**

B. Single-bond defect and periodic-chain comparison: two complementary topological changes

The two main experiments of this work — the single-bond defect scan and the periodic-chain comparison — represent two different types of topological changes.

The single-bond defect changes the local bond strength t , which is a local parameter in the Hamiltonian. The change in A (over 500%) was shown by the penetration depth measurement to be caused by the change in L_{eff} — a weak bond cuts off spin correlations, shortening the effective length. This phenomenon can be explained within the framework of the Kane–Fisher weak-link picture. The value of the single-bond defect experiment lies in: providing the first quantitative DMRG benchmark for

the variation of penetration depth with bond strength, confirming the Kane–Fisher picture in the Mott spin sector.

The periodic-chain comparison changes the boundary condition, which is a pure topological change — no local parameter is altered. The change in A (15%) cannot be attributed to “local parameter changed” or “poor choice of scaling variable” or “metal–insulator transition”. After excluding all these alternative explanations, the only remaining explanation is: the topological relation itself determines the physical response. The value of the periodic-chain comparison experiment lies in: it provides the first direct physical evidence for “relation precedes entity” — something the single-bond defect experiment could not provide.

The two experiments complement each other: the single-bond defect experiment shows how a local parameter change affects the global response through the penetration depth, and the periodic-chain comparison experiment shows how a pure topological change affects the physical response independently of local parameters. Together they form the complete empirical foundation for “relation precedes entity”.

C. Drahos’s three critical interventions

Every key step of this work has been advanced by Drahos’s technical contributions.

First intervention — identifying variable confounding: In an early $A(C)$ relation test, Drahos discovered through independent analysis of the raw data that C is a deterministic function of L on a linear chain ($C \approx 0.9745 - 0.624/L$, $R^2 = 0.99999$), so the high R^2 correlation between A and C was a statistical artifact, not a physical discovery. He proposed the correct multi-topology experimental scheme — fixing L and varying the topological structure. This correction redirected Prediction 1 from a dead end to the correct testing direction.

Second intervention — confirming the Kane–Fisher weak-link picture: In the single-bond defect scan experiment, Drahos pointed out that the huge variation of A with defect strength might simply be a change in the effective scaling length L_{eff} , not a failure of the bulk universal constant. He provided the exact Bethe ansatz values of the spin-wave velocity πv_s as an external standard, diagnosed the convergence boundary of $\chi_{\text{max}} = 100$, and cited the Kane–Fisher (1992) theory and related DMRG literature to correctly frame the physical origin of this phenomenon. This intervention revised the interpretation of Prediction 1 from a “new physical discovery” to “confirmation of the Kane–Fisher weak-link picture in the Mott spin sector”.

Third intervention — analytic confirmation of the periodic-chain Mott gap: In the periodic-chain comparison experiment, the first author failed to converge the charge gap calculation under periodic boundary conditions. Drahos used the exact Bethe ansatz to pro-

vide the Mott gap $\Delta_c(U=4) = 1.286727$, proving that it is analytically determined and boundary-independent, thereby closing the gate condition. He subsequently voluntarily corrected an earlier mistake of his own, distinguishing the removal energy $\mu_- = E(N) - E(N-1)$ from the true Mott gap Δ_c , clearly explaining the physical difference between $E(N) - E(N-1)$ and $E(N+1) + E(N-1) - 2E(N)$. This technical correction does not alter the physical conclusion of Experiment 2 in any way, but ensures the conceptual precision of this paper.

Drahoš's contributions to this work are constitutive, not auxiliary. Without his three interventions — identifying the variable confounding, positioning the Kane–Fisher picture, and analytically confirming the Mott gap — the conclusions of Experiments 1 and 2 could not have been presented in the current clear, honest, and referee-proof form.

VII. LIMITATIONS AND FUTURE DIRECTIONS

A. Current limitations

We honestly declare the known limitations of this work:

1. **Single parameter point:** The periodic-chain comparison experiment has so far been completed only at $U/t = 4$, $L = 40$. Tests at more U values and more L values are needed.
2. **Bond dimension limitation:** All DMRG calculations were performed at $\chi_{\max} = 100$, without multi- χ extrapolation.
3. **Single model:** The test has been completed only in the Fermi-Hubbard model. Tests in other strongly correlated models await future work.
4. **Independent verification:** Drahoš's independent $\chi \rightarrow \infty$ extrapolation verification of the periodic-chain A value has not yet been completed.

These limitations are not failures, but the honest boundaries of this work.

B. Future directions

1. Drahoš's independent DMRG verification: perform $\chi \rightarrow \infty$ extrapolation independent verification of the periodic-chain A value.
2. Multi- L test: supplement $L = 60$ periodic chain to confirm L -convergence.
3. Multi- U test: repeat the periodic-chain comparison experiment at other U values (e.g., $U = 1$, $U = 2$).

4. Multi-topology test: extend to other topological structures (e.g., ladder, two-dimensional square lattice) to test the universality of the topological boundary effect.

VIII. CONCLUSION

Through three independent DMRG experiments, this work has conducted a systematic study of the behavior of the spin-gap prefactor A in a one-dimensional Mott insulator. The core findings are as follows:

1. **Discovery of the topological boundary effect (core finding):** Under completely identical local entity parameters, simply changing the boundary condition (open $\rightarrow$ periodic) caused the spin-gap prefactor A to change from 3.0667 to 3.5262, a difference of about 15%. Drahoš confirmed the boundary-independence of the Mott gap using the exact Bethe ansatz, ruling out a transition. This is the first direct numerical evidence for “relation precedes entity” in strongly correlated physics. The standard theory prediction that A is a bulk universal constant independent of boundary conditions is negated.
2. **Penetration depth mechanism of the single-bond defect:** A varies monotonically with single-bond defect strength (ten data points, variation exceeding 500%). Penetration depth measurement reveals the physical origin: a weak bond cuts off spin correlations (penetration depth $\approx L/2$), and the effective scaling length L_{eff} replaces the geometric length L . This confirms the Kane–Fisher weak-link picture in the Mott spin sector, cleanly connecting the EDRN prefactor to L_{eff} for the first time.
3. **Universality of spin criticality:** The spin-gap prefactor $A = 5.497$ in the weak-coupling region ($U = 0.5$) is consistent with the Bethe ansatz exact value $\pi v_s = 6.028$, confirming that the Heisenberg universality class holds across the entire Mott phase diagram of the one-dimensional Hubbard model.
4. **Methodological demonstration:** This work demonstrates how a philosophical postulate can be converted into a testable numerical prediction and tested through rigorous decisive experiments in a concrete physical system. Every revision, every negation, and every re-positioning in the entire process has been fully documented and openly discussed.

The fundamental contribution of this work lies not in verifying a specific numerical value, but in proving that “relation precedes entity” is not merely philosophical speculation, but a scientific research program capable of producing testable predictions in strongly correlated physics. The 15% change in A in the periodic-chain

comparison experiment is the first empirical foundation of this program.

ACKNOWLEDGMENTS

All core DMRG calculations of this research were completed on a personal computer without the use of any institutional computing resources. We thank the developers of the TenPy library for providing the open-source DMRG tool.

We thank Rastislav Drahos for his decisive contributions to this work. He independently identified the variable confounding between C and L in the early experiment and proposed the correct fixed- L multi-topology experimental scheme; in the single-bond defect experiment, he used exact Bethe ansatz values to diagnose the convergence boundary of $\chi_{\max} = 100$ and correctly positioned the experimental finding as confirmation of the Kane–Fisher weak-link picture; in the periodic-chain comparison experiment, he used the exact Bethe ansatz to provide the analytic value of the Mott gap, closing the gate condition. From the initial identification of variable confounding to the final analytic confirmation, every one of his interventions corrected the research direction at a critical juncture. Without his contributions, this work could not have been completed in its current clear, hon-

est, and verifiable form.

We thank Marat Sultanov for performing cross-framework blind tests on multiple DMRG datasets using the TAT-Defense module. His “honest silence” on the charge compressibility data, the structural anchors detected on the spin gap and L-scan data, and his honest confirmation that the anchors were convergence artifacts after receiving precise diagnosis, established rigorous methodological standards for this collaboration. His TAT analysis is an irreplaceable part of the cross-framework verification.

We thank Qingkong for her diagnosis and propulsion of the collaborative network in Discussion #1466. Her pointing out that “the joint paper needs to be pushed forward” brought three independent lines of work to converge in the same direction. She did not directly produce physical data, but her organizational contribution enabled this collaboration to be completed.

All methodological decisions in this work — including open acceptance of external critique, withdrawal of flawed naming, side-by-side placement of data, and pre-publication of experimental protocols — strictly follow the core principles of rigorously preventing the tool-rationality paradox, firmly opposing alignment, and only diagnosing, not prescribing. These principles are not decorations, but the fundamental guarantee that this collaboration could arrive at the truth.

-
- [1] G. Li, R. Drahos, and M. Sultanov, *edrn-dmrg-verification* [Source code and experimental protocols], GitHub, <https://github.com/luoxuejian000/edrn-dmrg-verification> (2026).
 - [2] C. L. Kane and M. P. A. Fisher, *Transmission through barriers and resonant tunneling in an interacting one-dimensional electron gas*, Phys. Rev. B **46**, 15233 (1992).
 - [3] E. H. Lieb and F. Y. Wu, *Absence of Mott Transition in an Exact Solution of the Short-Range, One-Band Model in One Dimension*, Phys. Rev. Lett. **20**, 1445 (1968).
 - [4] I. Affleck, *Quantum spin chains and the Haldane gap*, J. Phys.: Condens. Matter **1**, 3047 (1989).
 - [5] T. Giamarchi, *Quantum Physics in One Dimension* (Oxford University Press, 2004).
 - [6] H. J. Schulz, *Correlation functions of one-dimensional correlated electron systems*, Phys. Rev. Lett. **64**, 2831 (1990).
 - [7] J. Hauschild and F. Pollmann, *Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy)*, SciPost Phys. Lect. Notes **5** (2018).